

MCT 333 Design of Mechatronics 1

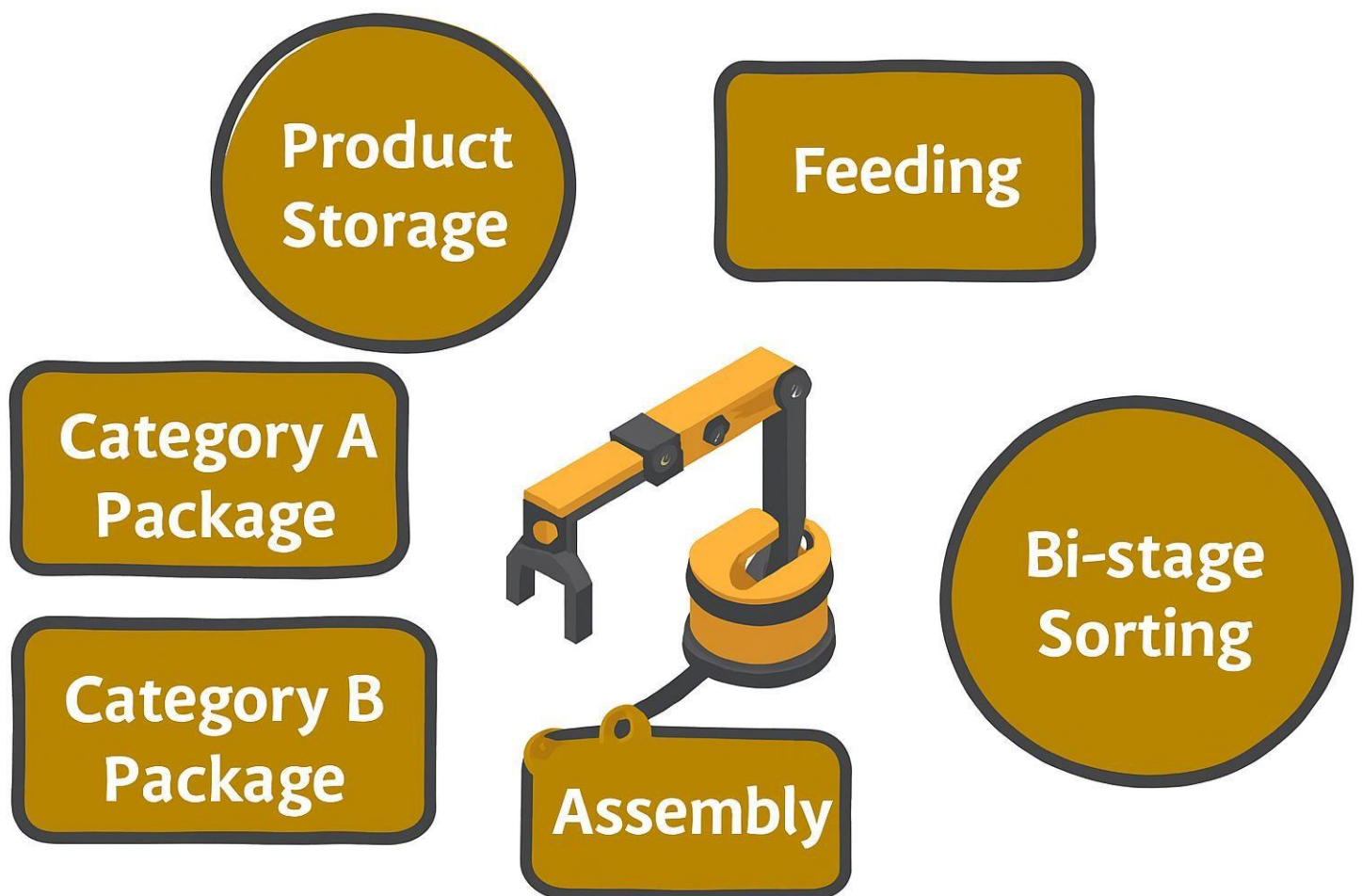
Fall 2026

Project Description



Project Objectives

The project should show students understanding the steps of the design of a mechatronic system on both a micro and macro cycle level as well as incorporating the product design parameters such as materials utilization, Human machine interface and modular approach of components. The stations integration and safety aspects will be evaluated as well as the successful testing of the product according to the requirements of each station



Project details

Every team needs to design a production line to be designed according to the Design of Mechatronics Systems standard VDI 2206. The main purpose of the production line is to handle cuboid or cylindrical shaped products and sort them based on shape then each sorted product category is packed into its package. The production line stages will be assigned to a group of **Five** students.

Project Requirements

- I. The Product will be consisted of two parts to be assembled. The design of the product is to be determined by the students
- II. The products should be stored at the beginning of the operation in either vertical or horizontal magazines, the products shouldn't be sorted in the magazines. Then it should be fed and transported to a bi-level sorting stage
 - First Sorting Stage: Sorting is performed based on either color, material, or weight.
 - Second Sorting Stage: Sorting is carried out according to product shape.
 - The following figures illustrate the overall concept of the sorting process.

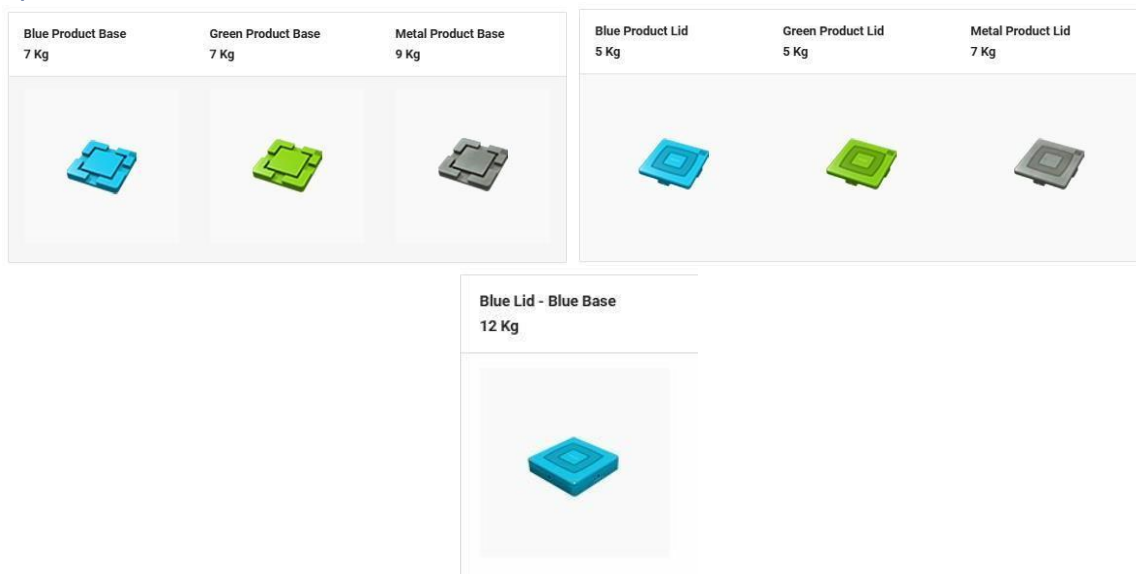
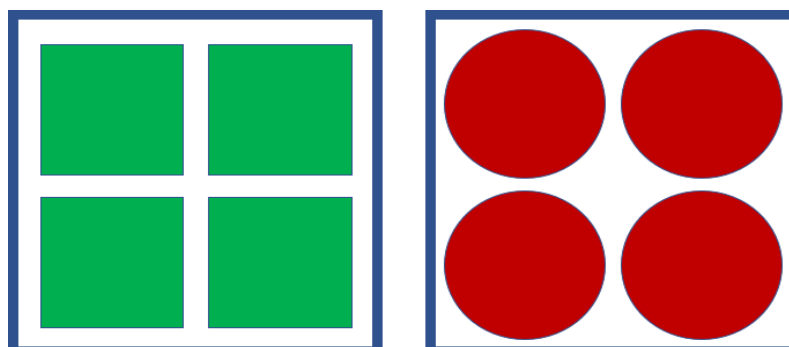


Figure 1 Final Product

- III. For the sorted products, the production line shall assemble and store them in a square shaped container such that 4 of them fit in one container as shown in the figure below. This storage area is designated exclusively for assembled products.



- IV. The storage system shall include two main storage areas and one temporary holding area. Bases must be stored first, followed by their corresponding lids. If a lid arrives before its base, it shall be placed in the temporary holding area until the base is available.
- V. The production line must have one or multiple **control boxes** that contains all electronic/electrical components {Micro-controllers, valves, Motor drivers...}, in addition the production line should provide a **User Interface** {e.g. LCD screen} that displays valid statistics to the user.
- VI. Implementation should also include **ESP-NOW communication** between the micro-controllers to accomplish any integration task. In addition, software development for each microcontroller doing several tasks should include usage of a **Real-time Operating system** to handle tasks scheduling.
- VII. The final implementation constrains is that the project must have a robotic arm with at least three degrees of freedom that can do several tasks or at least one task. At least one degree of freedom must be controlled using a PID Controller
- VIII. The project must meet all the requirements of mechatronic system.
- IX. Safety - Emergency Stop and Guarding: The production line shall include at least one hardware Emergency-Stop (E-Stop) that removes power from all actuators, and every exposed moving or pinch point shall be mechanically guarded.
- X. Fault Handling: The line shall detect a mis-fed, jammed, or faulty product and handle it safely (dedicated reject location or alarm) without stopping the whole line or causing a collision.
- XI. Sensing per Station: Each station shall include the sensors required to verify completion of its own task (e.g. part presence, position, or the specific sorting parameter it acts on).
- XII. Throughput / Cycle-Time Target: Each team shall define and meet a measurable throughput target (products per minute) and report the achieved cycle time for the full line.
- XIII. Modularity: Each station shall be mechanically and electrically modular using quick-connect interfaces so that it can be detached and tested independently.
- XIV. Live Statistics on the HMI: The User Interface shall display, as a minimum, live counts per product category, total throughput, current cycle time, and the active fault / E-Stop status.
- XV. Reliability and Repeatability: The complete production line shall run a defined number of consecutive product cycles autonomously, without manual intervention or jams, to be demonstrated at the Final submission.

Additional Technical Requirements

I. Electrical and Wiring

- The line shall have a single main power entry with a master switch and appropriate fusing / circuit protection on each supply rail.
- A common, clearly defined logic voltage (e.g. 5 V or 3.3 V) and a single star-ground reference shall be used across all boards.
- Motor and actuator power shall be separated from logic power, with flyback / snubber protection on all inductive loads.
- All wiring shall be routed inside the control box(es) with strain relief, labelled at both ends, and colour-coded per the wiring diagram.
- Connections between modular stations shall use keyed, lockable connectors (no bare wires between stations).

II. Software and RTOS

- Each microcontroller shall run an RTOS with a documented task table listing task name, priority, period, and stack usage.
- Periodic tasks shall use RTOS delays / timers; blocking busy-wait delays are not permitted.
- Inter-task communication shall use queues, semaphores, or event flags rather than unprotected shared globals.
- A watchdog timer shall be enabled and shall recover the controller into a safe state on a hang.
- ESP-NOW messages shall carry an identifier and defined payload, and shall be acknowledged and retried on failure.
- Each controller shall implement defined safe states (E-Stop pressed, communication lost) and shall fail safe.

III. Control and PID

- The PID-controlled degree of freedom shall run at a fixed, documented sampling rate.
- The controller shall include anti-windup on the integral term and output saturation limits.
- The tuning method and final gains (K_p , K_i , K_d) shall be documented.
- The controlled axis shall meet defined targets for overshoot, steady-state error, and settling time, supported by evidence.
- Sensor feedback used for control shall be filtered / conditioned as needed and its resolution stated.

IV. Mechanical and Design-for-Manufacture

- Load-bearing joints shall be fastened (bolted / screwed), not glued.
- Parts shall use standard stock sizes and standard fasteners; custom dimensions shall be minimised.
- Mating parts shall have defined tolerances / fits and alignment features (dowels, slots, locating tabs) where accuracy matters.
- Assemblies shall be serviceable: fasteners and adjustment points accessible without full disassembly.
- Moving parts shall have adequate clearance and shall use guides or bearings rather than sliding plastic-on-plastic where wear matters.

Design Constraints

Design Constraints (Cost-Driven): The following constraints are mandatory and are intended to drive the manufacturing cost down. Compliance will be assessed in the Major Task and Final submissions.

1. 3D-Printing Limit: 3D-printed parts shall not exceed 15% of the total manufacturing. Every printed part must be justified by a function or geometry that cannot reasonably be achieved by cheaper means.
2. Off-the-Shelf First: Standard, off-the-shelf, and stock components (e.g. aluminium / T-slot extrusion, standard fasteners and bearings, laser-cut acrylic / MDF / plywood) shall be preferred over custom-machined or fully 3D-printed structures.
3. Minimise Total Cost: There is no fixed budget; however, teams shall keep the total Bill of Materials (BOM) cost as low as possible. Every design decision shall favour the lowest-cost option that still meets the requirement, and the achieved BOM cost will be part of the evaluation.
4. Cost Log: Teams shall maintain a live BOM with unit prices and a running total and submit an as-built cost sheet with the Major Task.
5. Right-Sized Actuators: Actuators shall be sized to the load (see Actuator Sizing); over-specified or exotic actuators shall be avoided to reduce cost.
6. Standardisation: Teams shall standardise on a single microcontroller family and a minimal set of fastener and stock sizes to reduce cost and spare-part count.
7. Material-Waste Minimisation: Sheet and raw-stock parts shall be nested and laid out to minimise material waste.

General Submission Instructions

- 1) A Technical Report That meets all the report writing guidelines (e.g. Table of Content, List of Figures, Justified Text, Properly Captioned Figures, etc)
- 2) A Contribution Table detailing the specific tasks completed by each member (Percentages are not acceptable)
- 3) An Oral Discussion will be held during each submission; therefore, every student is expected to have a thorough understanding of all the work presented.

Quiz Submission (5 Marks)

Write a Project Proposal that should incorporate the following

- Brief Description about the main Idea your project
- Subsystem Design Sketches
- Integration Ideas with sketches
- Work Break Down Structure
- Gantt Chart
- Suggested BOM with reasoning behind Key components

Acceptance Criteria:

- Main idea, subsystems, and integration approach are clearly explained with sketches.
- A complete Work Breakdown Structure and a dated Gantt chart covering all milestones are included.
- A preliminary BOM lists key components with justification and estimated costs.
- Each team member's responsibilities are identified.

Midterm Submission (25 Marks)

1) CAD Design

- Full CAD Design with all the mechanical consideration are met
- A Fully Documented Manufacturing Plan and Material Selection
- Full Actuator Sizing for any present actuator
- CAD Files

2) Simulation

- Fully Functional Simulation for at least 2 Products
- Contact Forces Toolbox Must be used
- System control is expected to be implemented using Stateflow
- Sensing from contact forces is to be incorporated into the Stateflow logic
- Simulation Files

Acceptance Criteria:

- Full CAD assembly is complete, fits together, and reflects the intended manufacturing method and materials.
- Actuator sizing calculations are shown for every actuator and support the selected components.
- Simulation runs successfully for at least two products using the Contact Forces toolbox.
- Stateflow logic controls the process and reacts to sensed contact forces.
- Manufacturing plans and material selection are documented and consistent with the Design Constraints.

Major Task Submission

The Major Task consists of two milestones (15 marks each, 30 total), both focused on ONE complete subsystem of the production line. Milestone 1 delivers that subsystem fully manufactured; Milestone 2 delivers the same subsystem fully functional in stand-alone operation. Integrating all subsystems into the complete line is assessed later in the Final submission.

Milestone 1 - A Complete Manufactured Subsystem (15 Marks)

1. A fully manufactured subsystem, identical to the CAD design, with all mechanical parts produced and assembled.
2. Fully Connected Electrical System with proper cable management with its Wiring Diagram
3. Stress Analysis on critical components by means of any selected software
4. As-built Bill of Materials with unit costs and running total, demonstrating compliance with the Design Constraints (3D-printing $\leq 15\%$, off-the-shelf-first, lowest-cost).
5. Implemented and verified safety features: a functional Emergency-Stop and guarding of all moving / pinch points.

Acceptance Criteria:

- The manufactured subsystem matches the CAD design dimensionally and in layout.
- The subsystem electrical system is fully wired, powered safely, and matches the submitted wiring diagram.
- Cable management is tidy, labelled, and strain-relieved.
- Stress analysis identifies critical components and shows acceptable safety factors.
- The as-built BOM demonstrates 3D-printed content $\leq 15\%$ and justifies component choices on cost.
- The Emergency-Stop removes actuator power and guarding is in place.

Milestone 2 - A Complete Functional Subsystem (15 Marks)

1. Software Flow chart for every microcontroller available shows the following:
 - RTOS Tasks
 - How Communication is handled between two microcontrollers
 - PID Control Algorithm
2. A Fully Automated Activity based Subsystem
3. The subsystem runs its complete activity cycle automatically and repeatably in stand-alone operation (not yet integrated with the other stations).
4. PID performance evidence: measured step-response / tracking plots for the PID-controlled degree of freedom.
5. A working HMI for the subsystem displaying its live statistics (e.g. product counts, cycle time, and status).
6. Acceptance test of the functional subsystem against its own requirements, with a recorded results log.

Acceptance Criteria:

- Every microcontroller in the subsystem runs its RTOS tasks per the documented task table.
- Any communication used within the subsystem is reliable, with acknowledgement / retry.
- The PID-controlled axis meets its stated overshoot, steady-state error, and settling targets, with evidence.
- The subsystem completes its full activity cycle automatically and repeatably on its own.
- The subsystem HMI shows live, correct statistics.
- The acceptance-test log records results against the subsystem requirements.

Final Submission (40 Marks)

The Final submission is worth 40 marks and evaluates the complete, fully integrated, and reliably operating production line - all subsystems working together as one - with the comprehensive final report and oral defense. It goes beyond the single subsystem delivered in the milestones.

1. Live demonstration of the complete production line running autonomously through a full cycle (feeding, bi-level sorting, assembly, and storage) for multiple products of both shapes.
2. Full integration of all station subsystems operating together without manual intervention, communicating over ESP-NOW.
3. The robotic arm (at least three degrees of freedom, at least one PID-controlled joint) performing its task(s) within the integrated cycle.
4. A complete HMI displaying live statistics for the whole line.
5. A comprehensive final technical report covering the full VDI 2206 design cycle and consolidating all previous submissions, reflecting the as-built system.
6. A Requirement Verification Matrix mapping every Project Requirement, Additional Technical Requirement, and Design Constraint to its evidence of compliance.
7. A final Bill of Materials with actual costs, confirming compliance with the Design Constraints (3D-printing $\leq 15\%$, off-the-shelf-first, lowest cost).
8. A short video recording of the system in full operation.

Acceptance Criteria:

- The full line completes the required number of consecutive product cycles autonomously, without intervention or jams.
- Both sorting levels classify correctly for cuboid and cylindrical products.
- Assembly and storage follow the base-then-lid rule and use the temporary holding area correctly.
- ESP-NOW communication, RTOS scheduling, and PID control operate reliably throughout the full run.
- Safety systems (Emergency-Stop and guarding) function during live operation.
- Every entry in the Requirement Verification Matrix is demonstrated or justified.
- The final report meets all report-writing guidelines and reflects the as-built system.